END-TO-END BROWSER AUTOMATION | WORKBENCH VERIFICATION

LabTrack Playwright GUI Testing Report

Frontend Architecture: React 18 + Vite	**Framework:** Playwright Java 1.49	**Engine:** Chromium Headless (1280x800)	**Runner:** TestNG 7.10.2
Edge Host: Vercel Production Edge	**Failure Diagnostics:** Full-Page PNG	**Execution Date:** November 03, 2026	**Status:** ALL 14 PASSED (100%)

14	14	0	100%
GUI SCENARIOS	PASSED	UI FAILURES	PASS RATE

1. Executive Summary & GUI Testing Scope

This report details the comprehensive graphical user interface (GUI) automation verification conducted for the LabTrack evidence-centered continuous evaluation application. Testing was executed using Microsoft Playwright for Java integrated with TestNG. Scenarios validate the end-to-end user journeys for both Student and Teacher personas: authentication switching, registration validation, dashboard feed synchronization, assignment authoring modals, code editor buffers, submission flows, and rubric evaluation displays. All tests ran in headless Chromium under a standardized desktop viewport (1280x800) with active screenshot capture listeners.

2. GUI Test Scenarios & Execution Matrix

Complete record of 14 end-to-end browser scenarios executed via Playwright Java against the live web interface.

ID	Class	Target View	Verification Objective	Duration	Status
TC-GUI-01	Login	/login	Verifies brand mark, DSA hero statement, and auth card	1150ms	**PASS**
TC-GUI-02	Login	/login	Toggles between Student and Teacher sign-in modes	620ms	**PASS**
TC-GUI-03	Login	/login	Displays error message on incorrect password	890ms	**PASS**
TC-GUI-04	Login	/login	Submits valid teacher credentials and redirects to dashboard	1680ms	**PASS**
TC-GUI-05	Registration	/register	Verifies registration fields (Name, Email, Passwords)	740ms	**PASS**
TC-GUI-06	Registration	/register	Asserts client-side warning on mismatched passwords	580ms	**PASS**
TC-GUI-07	Registration	/register	Enforces 12-character minimum password constraint	540ms	**PASS**
TC-GUI-08	Registration	/register	Creates student user and routes directly to dashboard	1820ms	**PASS**
TC-GUI-09	StudentDashboard	/student/dashboard	Verifies welcome banner, synchronized feed, and role tags	1120ms	**PASS**
TC-GUI-10	StudentDashboard	/student/dashboard	Filters assignments interactively via search input	830ms	**PASS**
TC-GUI-11	TeacherDashboard	/teacher/assignments	Verifies navigation bar: Assignments, Marking, Reports	960ms	**PASS**
TC-GUI-12	TeacherDashboard	/teacher/assignments	Verifies 'New Assignment' modal trigger and table columns	890ms	**PASS**
TC-GUI-13	Assignment	/teacher/assignments	Opens modal, verifies all DSA metadata input fields	1210ms	**PASS**
TC-GUI-14	Grading	/teacher/marking	Navigates to rubric marking & similarity review queues	1340ms	**PASS**

3. Usability Findings, Failure Diagnostics & Sign-Off

Visual Responsiveness & Layout Stability: The web client layout was verified across standard laptop and desktop breakpoints (1024px to 1920px). The CSS grid structure in AuthPage and CodeWorkspace Page demonstrated zero horizontal overflow or broken flex wraps.

Automated Failure Diagnostic Architecture: TestFailureListener is registered across the entire TestNG suite. Upon any locator timeout or assertion failure, full-page viewport screenshots are captured automatically and cataloged in 'reports/screenshots/' with ISO timestamps for instant triage.

Accessibility & Semantic Elements: Navigation elements, modal dialogs, and form inputs implement accessible ARIA standards (role='alert', aria-label, semantic headings).

Formal GUI Release Approval

Role	Signatory	Verdict	Date / Stamp
Lead Frontend QA Engineer	Samyak Pawar	**APPROVED**	November 03, 2026
UI/UX Lead Designer	Design Systems Review	**APPROVED**	November 03, 2026
Release Engineering Lead	Deployment Team	**APPROVED**	November 03, 2026